Yulu Electric Cycle Demand Analysis

Problem Statement

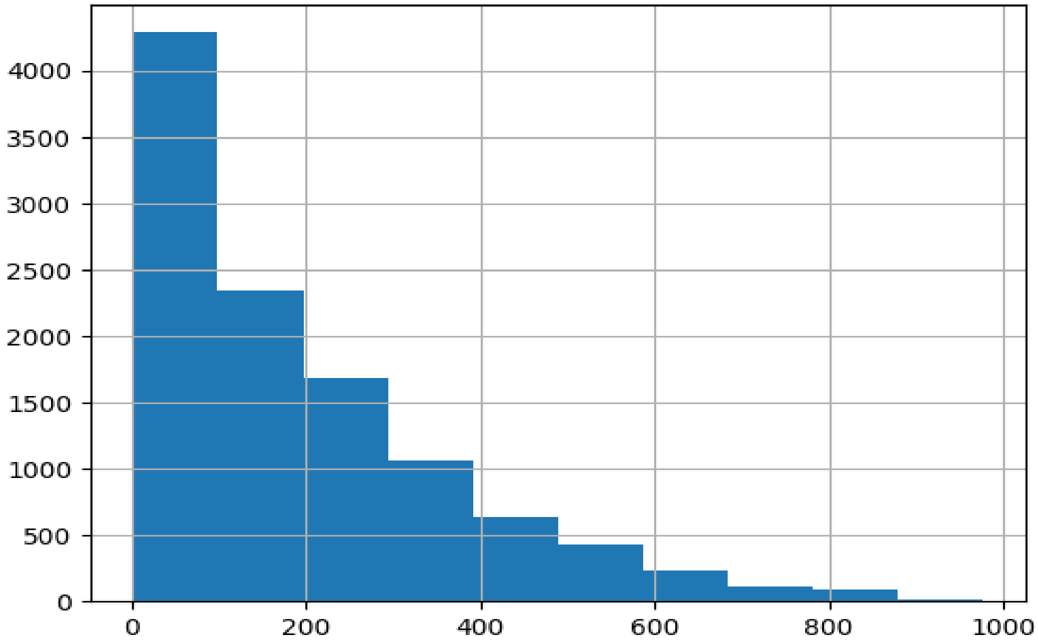
Identify key factors influencing demand for shared electric cycles.

Dataset Overview

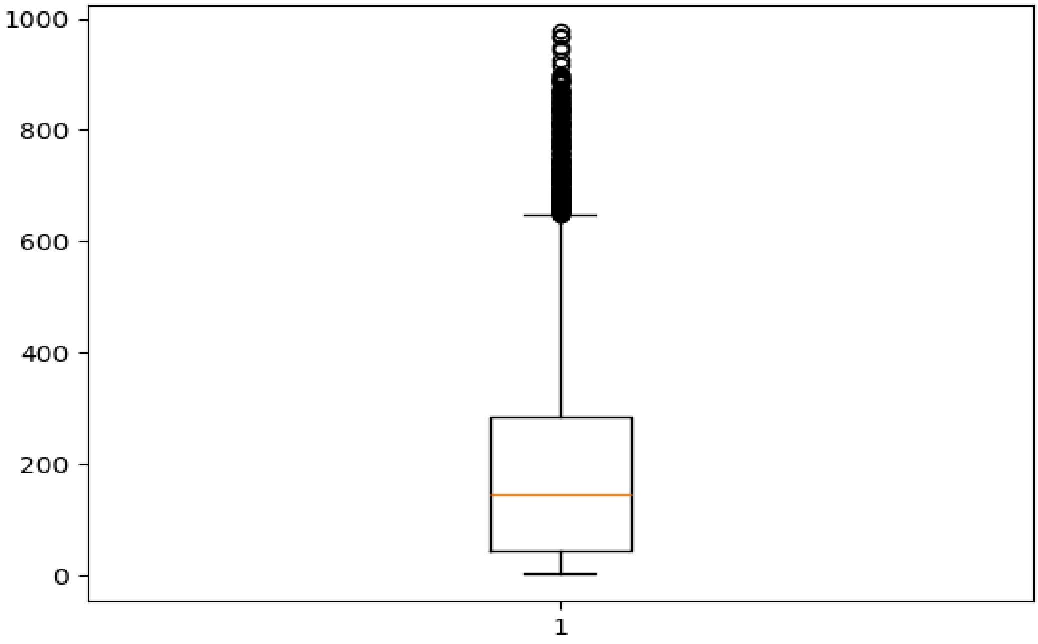
Shape: (10886, 13)

datetime season ... count hour count 10886 10886.000000 ... 10886.000000 10886.000000 mean 2011-12-27 05:56:22.399411968 2.506614 ... 191.574132 11.541613 min 2011-01-01 00:00:00 1.000000 ... 1.000000 0.000000 25% 2011-07-02 07:15:00 2.000000 ... 42.000000 6.000000 50% 2012-01-01 20:30:00 3.000000 ... 145.000000 12.000000 75% 2012-07-01 12:45:00 4.000000 ... 284.000000 18.000000 max 2012-12-19 23:00:00 4.000000 ... 977.000000 23.000000 std NaN 1.116174 ... 181.144454 6.915838 [8 rows x 13 columns]

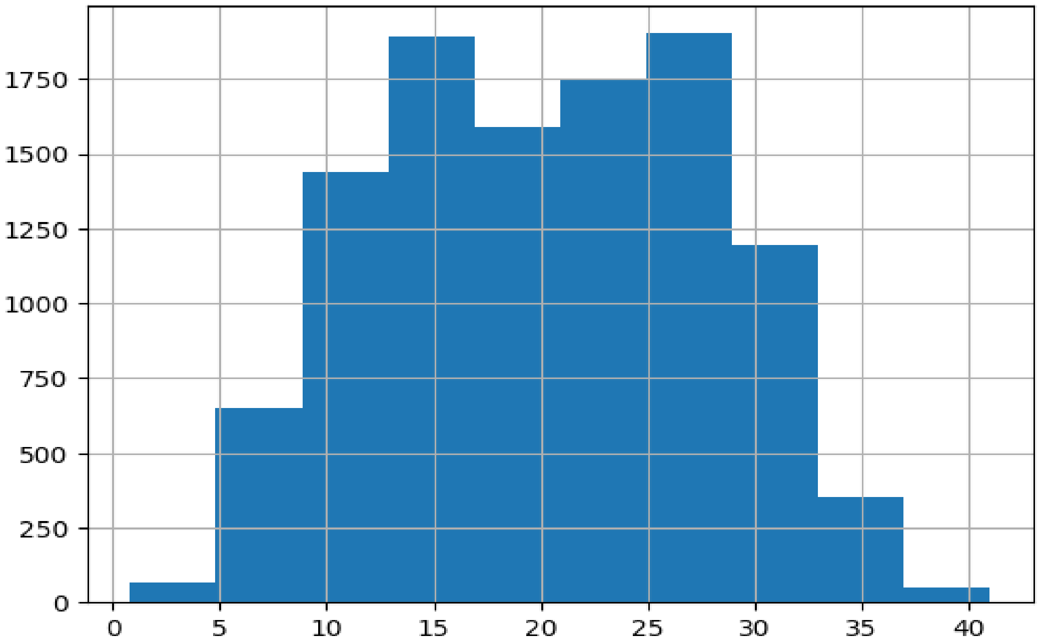
Univariate Analysis



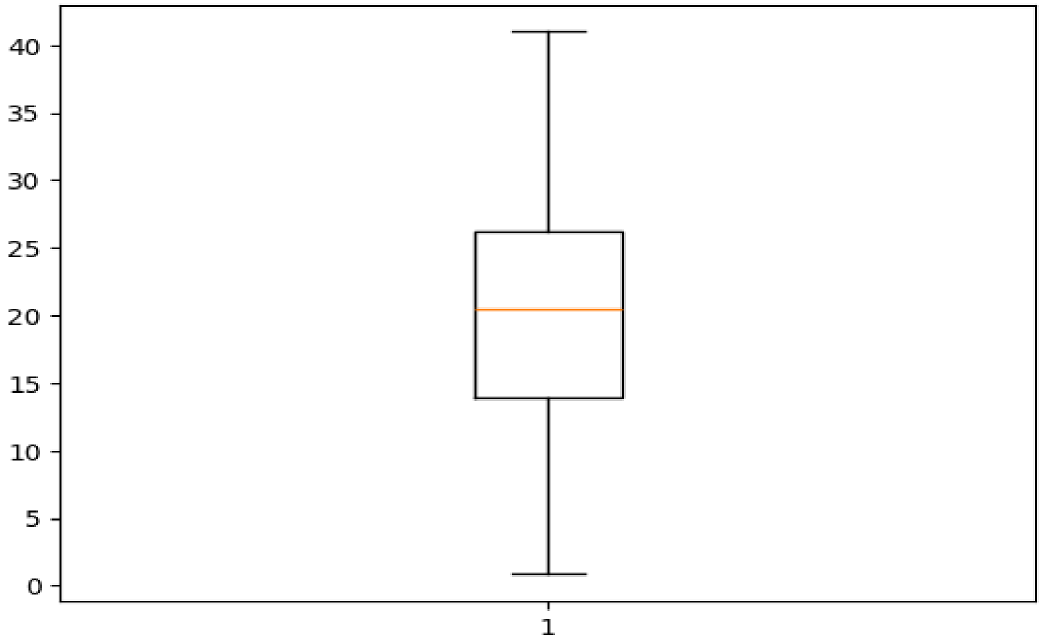
Distribution shows variability and skewness.



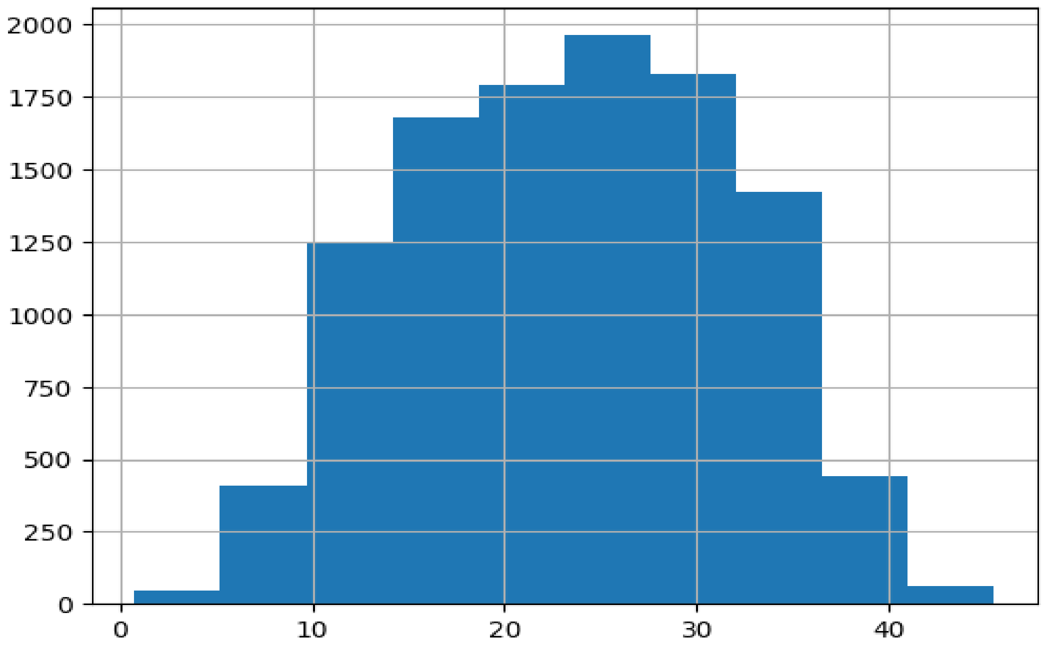
Boxplot shows spread and outliers.



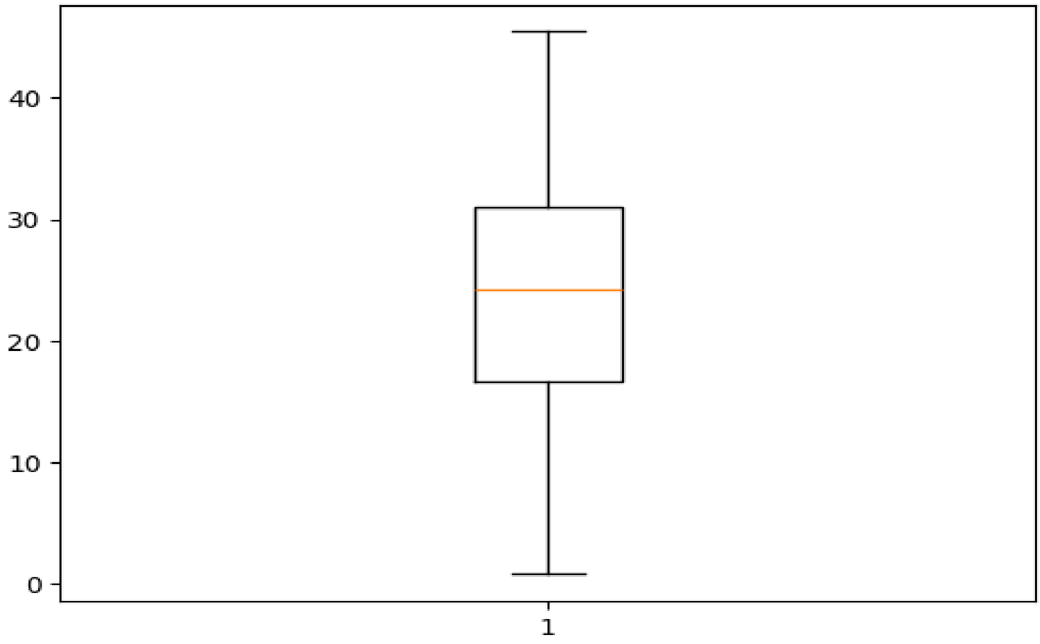
Distribution shows variability and skewness.



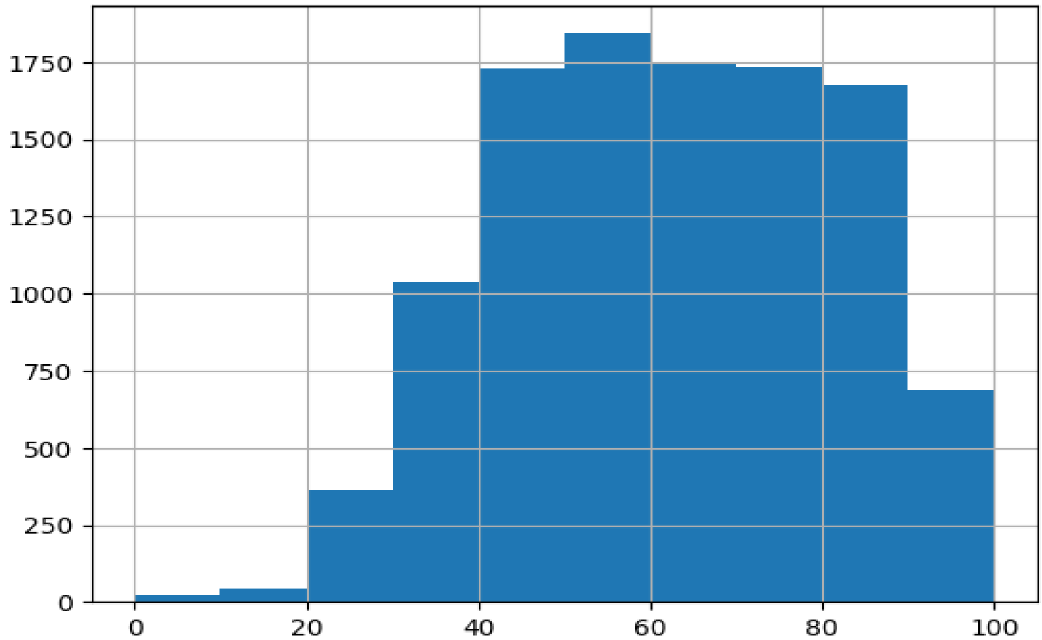
Boxplot shows spread and outliers.



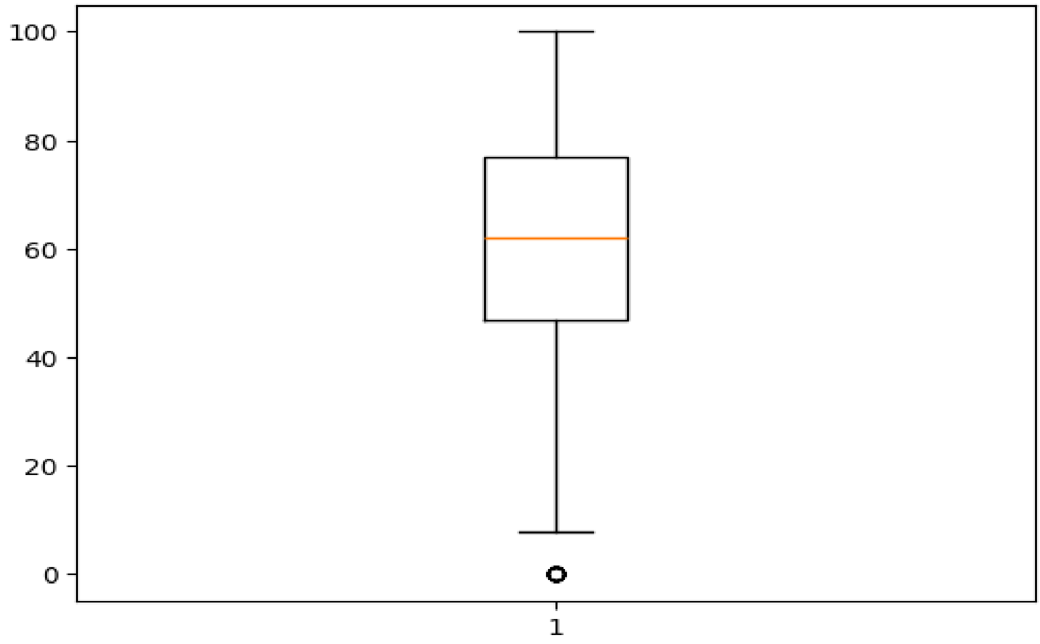
Distribution shows variability and skewness.



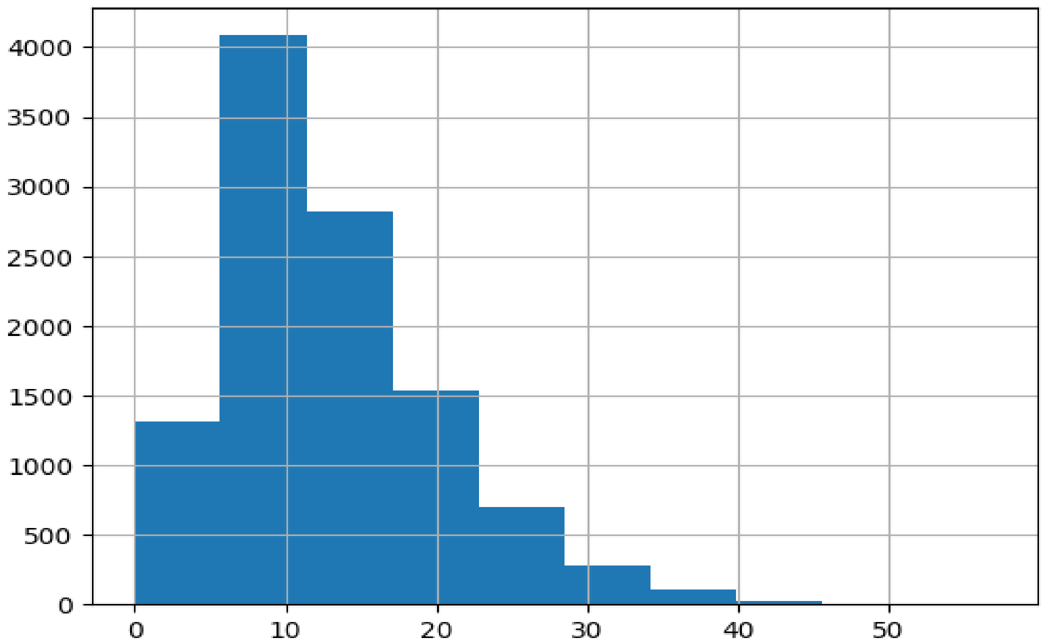
Boxplot shows spread and outliers.



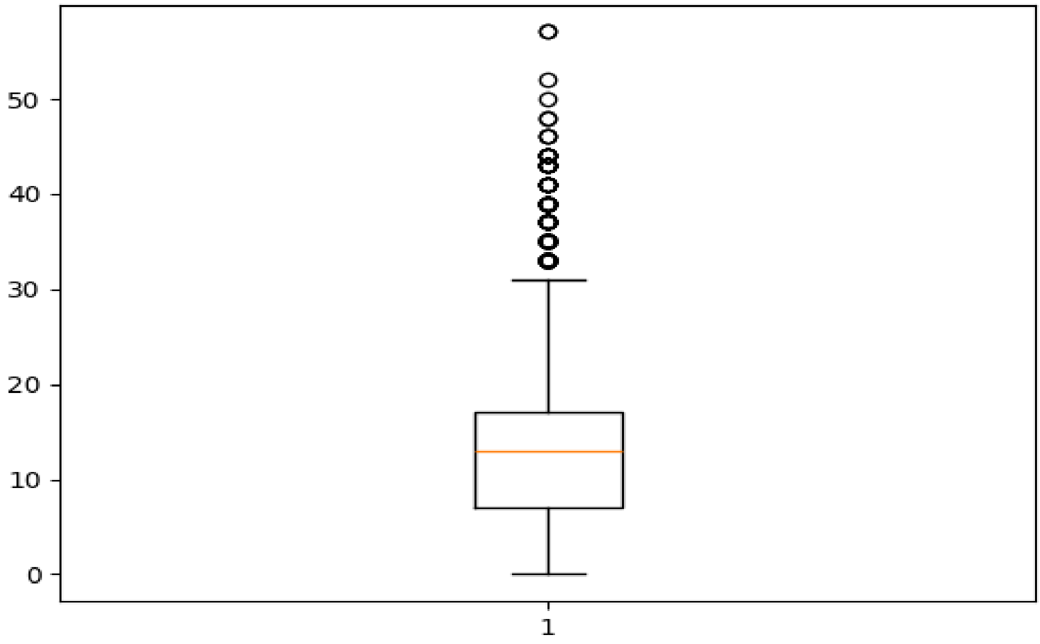
Distribution shows variability and skewness.



Boxplot shows spread and outliers.

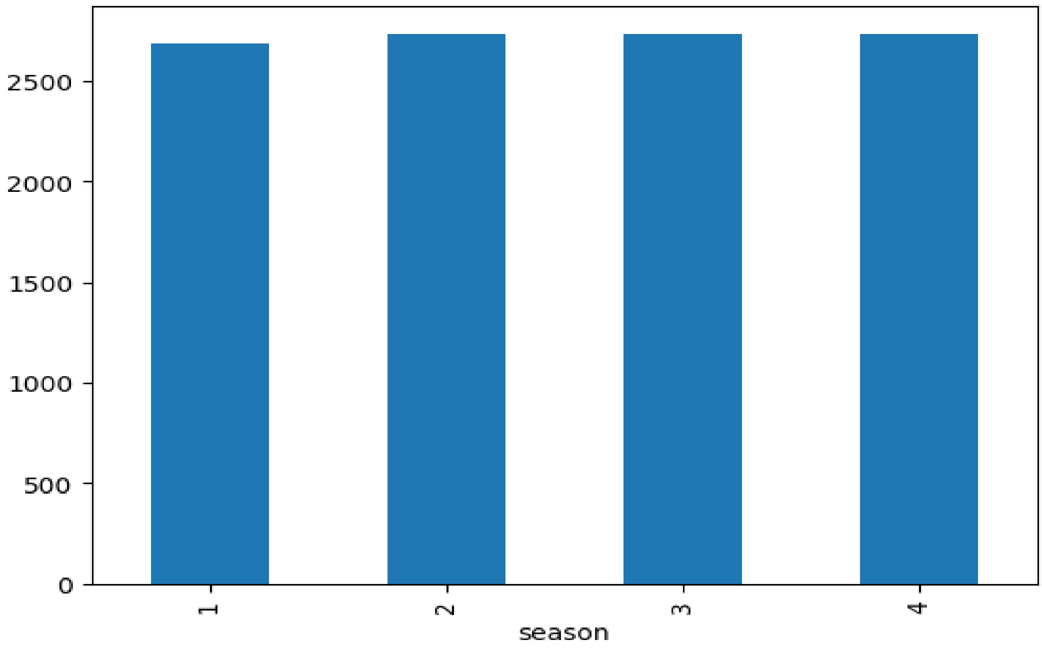


Distribution shows variability and skewness.

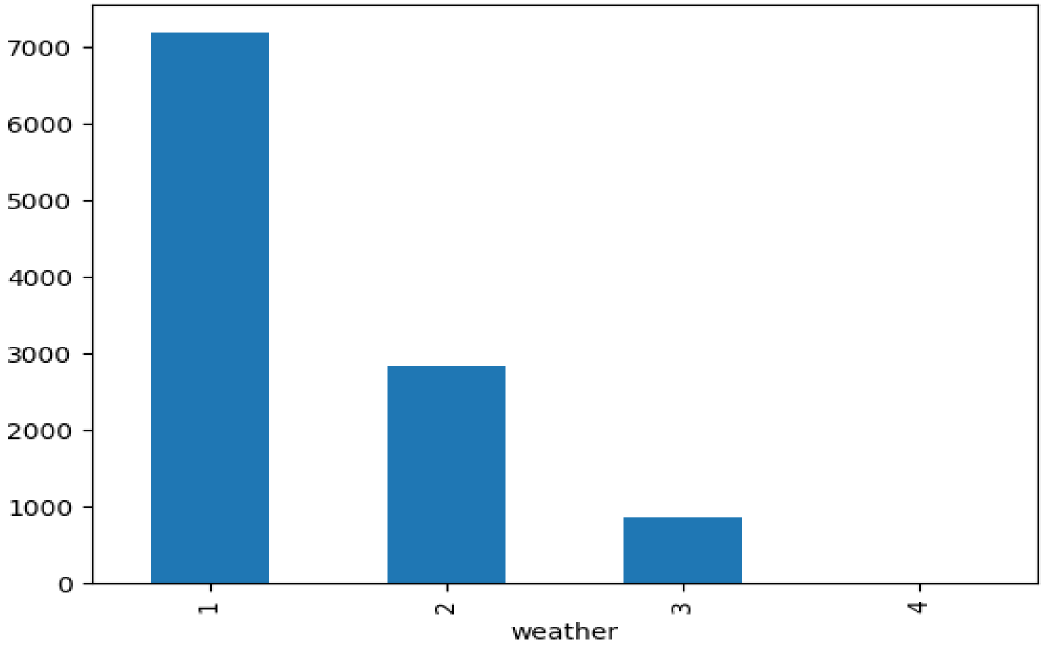


Boxplot shows spread and outliers.

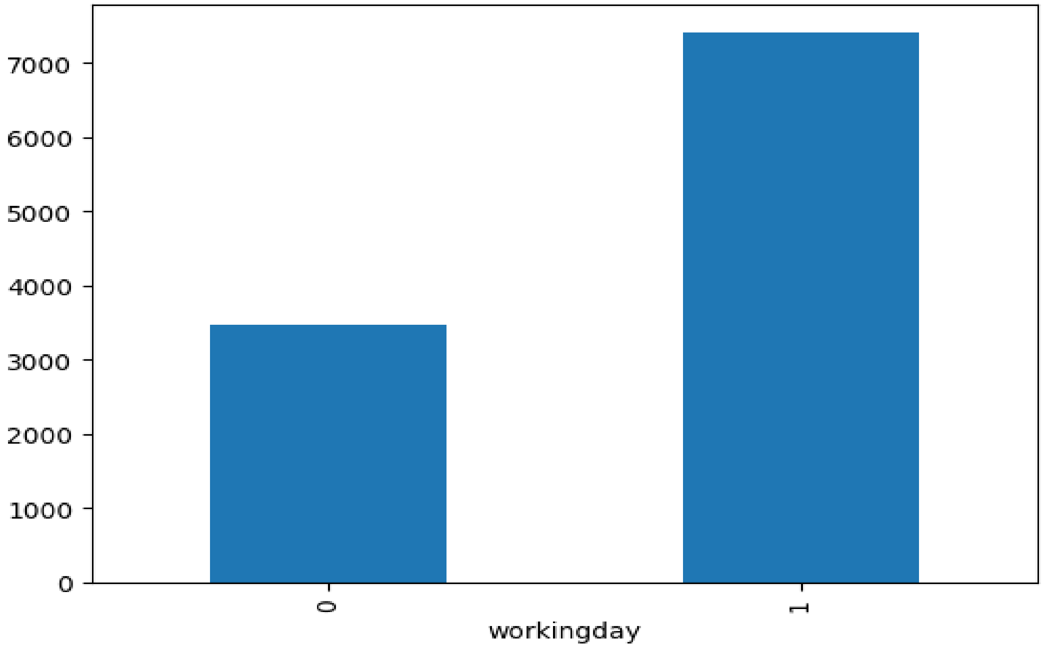
Categorical Analysis



Category distribution indicates imbalance or dominance.

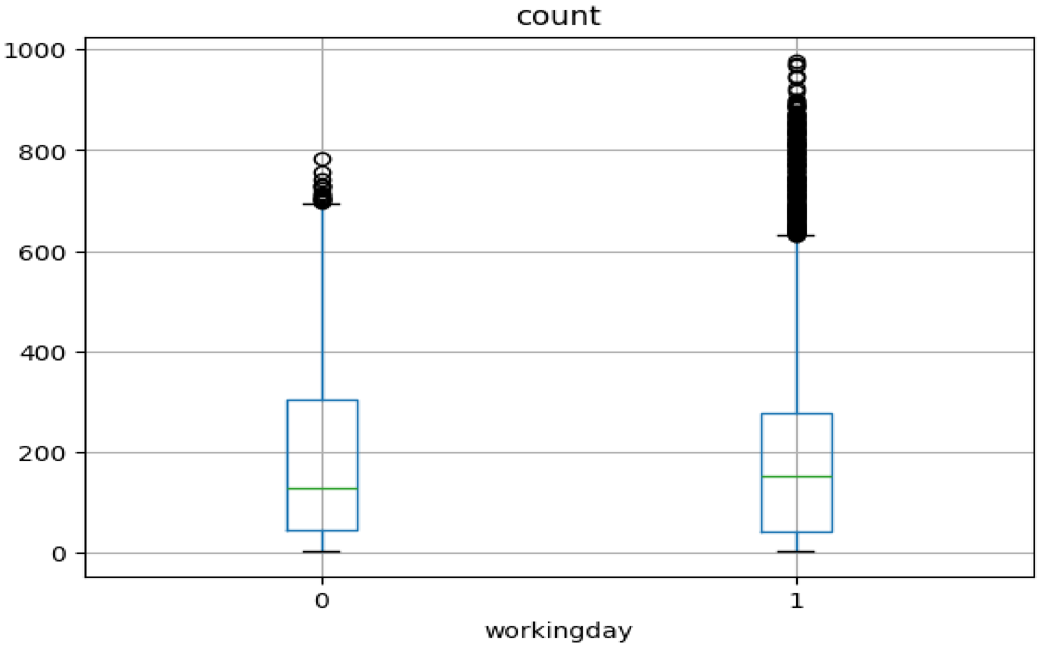


Category distribution indicates imbalance or dominance.

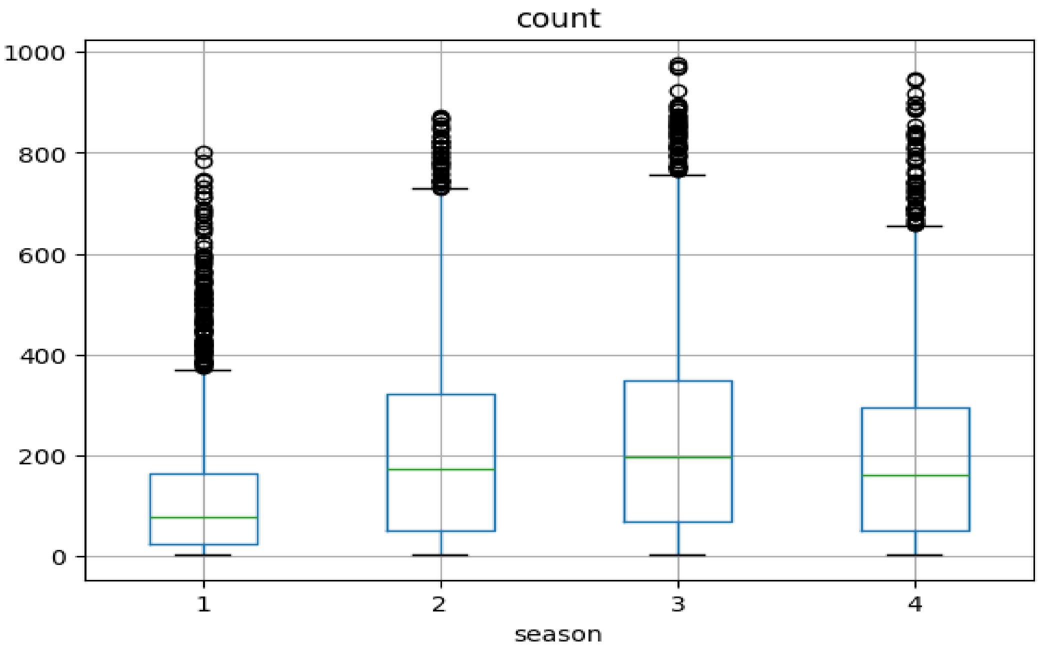


Category distribution indicates imbalance or dominance.

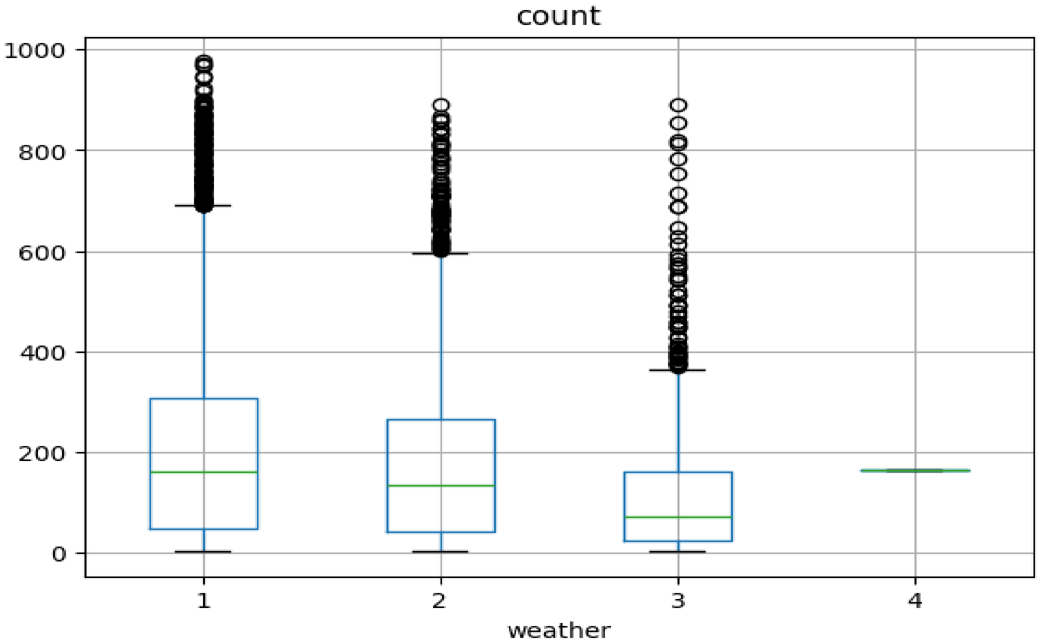
Bivariate Analysis



Boxplot shows variation in demand across categories.



Boxplot shows variation in demand across categories.



Boxplot shows variation in demand across categories.

Hypothesis Testing

T-test: t=1.24, p=0.21640

ANOVA Season: F=236.95, p=0.00000

ANOVA Weather: F=65.53, p=0.00000

Chi-square: chi2=49.16, p=0.00000

Final Insights

Demand depends on weather, season, and working day patterns.

Business Recommendations

Increase availability during peak demand periods and optimize pricing.